Pharmaceutical Calculations 2

Conversions Between Strengths

1. Convert the following into the given unit

a. 1 in 200 → 1 g : 200 ml → 1000 mg : 200 ml → 10 mg : 2 ml → 5 mg : 1 ml → **5 mg/ml**

b. 1 in 400 → 1 g : 400 ml → 1000 mg : 400 ml → 10 mg : 4 ml → **2.5 mg/ml**

c. 1 in 25 → 1000 mg : 25 ml → 200 mg : 5 ml → 40 mg : 1 ml → **40 mg/ml**

d. 1 in 1000 → 1 g : 1000 ml → 0.1 g : 100 ml → **0.1 %w/v**

e. 1 in 500 → 1 g : 500 ml → 0.1 g : 50 ml → 0.2 g : 100 ml → **0.2 %w/v**

f. 1 in 5000 → 1 g : 5000 ml → 0.01 g : 50 ml → 0.02 g : 100 ml → **0.02 %w/v**

g. 1:10000 → 1 g : 10000 ml → 1000 mg : 10000 ml → 0.1 mg : 1 ml → **0.1 mg/ml**

h. 1:250 → 1 g : 250 ml → 4 g : 1000 ml → 0.4 g : 100 ml → **0.4 %w/v**

i. 500 mg/ml → 500 mg : 1 ml → 0.5 g : 1 ml → 1 g : 2 ml → **1 in 2 (1:2)**

j. 125 mg/ml → 125 mg : 1 ml → 250 mg : 2 ml → 1000 mg : 8 ml → 1 g : 8 ml → **1 in 8 (1:8)**

k. 5% w/v → 5 g : 100 ml → 1 g : 20 ml → **1 in 20 (1:20)**

l. 0.2% w/v → 0.2 g : 100 ml → 2 g : 1000 ml → 1 g : 500 ml → **1 in 500 (1:500)**

Calculations with Strengths

2. If 25 g of a chemical is dissolved in 100 ml, what is this expressed in parts?

   25 g : 100 ml → 1 g : 4 ml → **1 in 4 (1:4)**

3. If 80 g of a chemical is dissolved in 400 ml, what is this expressed in parts?

   80 g : 400 ml → 8 g : 40 ml → 1 g : 5 ml → **1 in 5 (1:5)**

4. If 500 mg of a chemical is dissolved in 200 ml, what is this expressed in %w/v?

   0.5 g : 200 ml → 0.25 g : 100 ml → **0.25 %w/v**

5. If 1200 mg of a chemical is dissolved in 1 l, what is this expressed in %w/v?

   1.2 g : 1000 ml → 0.12 g : 100 ml → **0.12 %w/v**

6. If a formula is described as 1:500, how may ml are required to provide a dose of 25 mg?

   1 g : 500 ml → 1000 mg : 500 ml → 100 mg : 50 ml → 25 mg : 12.5 ml → **12.5 ml**

7. If a formula is described as 1:2000, how many ml are required to provide a dose of 100 mg?

   1 g : 2000 ml → 1000 mg : 2000 ml → 100 mg : 200 ml → **200 ml**

8. If a formula is 2% w/v, how many ml are required to provide a dose of 50 mg?

   2 g : 100 ml → 2000 mg : 100 ml → 200 mg : 10 ml → 100 mg : 5 ml → 50 mg : 2.5 ml → **2.5 ml**

9. If a formula is 10% w/v, how many ml are required to provide a dose of 40 mg?

   10 g : 100 ml → 10000 mg : 100 ml → 10 mg : 0.1 ml → 40 mg : 0.4 ml → **0.4 ml**

10. If 200 ml of water is added to 800 ml of a 1 in 200 solution, what is the final strength:

    Original 1 g : 200 ml → 4 g : 800 ml

    Add 200 ml of water so you now have 4 g : (800 + 200) ml → 4 g : 1000 ml

    a. Expressed in parts?

       4 g : 1000 ml → 1 g : 250 ml → **1 in 250**

    b. Expressed in %v/v?

       4 g : 1000 ml → 0.4 g : 100 ml → **0.4 %w/v**

11. If 50 ml of water is added to 200 ml of a 1:400 solution, what is the final strength:

    Original 1 g : 400 ml → 0.5 g : 200 ml

    Add 50 ml of water, so you now have 0.5 g : 250 ml

    a. Expressed in parts?

       0.5 g : 250 ml → 1 g : 500 ml → **1 in 500**

    b. Expressed in %w/v?

       0.5 g : 250 ml → 1 g : 500 ml → 0.2 g : 100 ml → **0.2 %w/v**

12. If 500 ml of water is added to 2 l of a 20% w/v solution, what is the final strength expressed as %w/v?

Original 20 g : 100 ml → 400 g : 2000 ml (2000 ml is 2 l)

Add 500 ml of water : 400 g : 2500 ml

400 g : 2500 ml → 4 g : 25 ml → 16 g : 100 ml → **16 %w/v**

13. If 400 ml of water is added to 1 l of a 14% w/v solution, what is the final strength expressed as %w/v?

    Original 14 g : 100 ml → 140 g : 1000 ml

    Add 400 ml of water: 140 g : 1400 ml

    140 g : 1400 ml → 10 g : 100 ml → **10% w/v**

Dilutions

14. If you have stock of a medication that is given as 1 in 100, and you need to create 40 ml of a 1 in 200 solution, how many ml do you need to take from the concentrate?
    1 in 200 is half the strength of 1 in 100, so take **20 ml**
    Or

$$C1 = \frac{1}{100} = 0.01, V1 = ?, C2 = \frac{1}{200} = 0.05, V2 = 40$$

$$C1V1 = C2V2 \rightarrow 0.1V1 = 0.05 \times 40$$
$$V1 = \frac{0.05 \times 40}{0.01} = \frac{0.2}{0.01} = \mathbf{20\ ml}$$

15. If you have a stock of medication that has strength 1 in 200, and you want to make 100 ml of a 1 in 1000 solution, how many ml do you need to take from the concentrate?
    1 in 1000 is a 5 times dilution of 1 in 200, so need 100/5 = **20 ml**

16. If you want to make 200 ml of a medication that is 5% v/v, and you have stock medication that is 25% v/v, how much of the stock would you take?

$$C1 = 25, V1 = ?, C2 = 5, V2 = 200$$
$$C1V1 = C2V2$$
$$25V1 = 5 \times 200$$
$$V1 = \frac{5 \times 200}{25} = \frac{1000}{25} = \mathbf{40\ ml}$$

17. How much of a 5 molar solution is required to produce 200 ml of a 1 molar solution?

$$C1 = 5, V1 = ?, C2 = 1, V2 = 200$$

$$C1V1 = C2V2$$
$$5V1 = 1 \times 200$$
$$V1 = \frac{1 \times 200}{5} = \frac{200}{25} = \mathbf{40\ ml}$$

18. How much of a 4 molar solution is required to produce 500 ml of a 500 millimolar solution?

$$C1 = 4, V1 = ?, C2 = 0.5, V2 = 500$$
$$C1V1 = C2V2$$
$$4V1 = 0.5 \times 500$$
$$V1 = \frac{0.5 \times 500}{4} = \frac{250}{4} = \mathbf{62.5\ ml}$$

19. If 100 ml of compound X in a 4 molar solution is required to react completely with 20 ml of compound Y, what is the strength of compound Y's solution, assuming the reaction was 1X:1Y?

$$C1 = 4, V1 = 100, C2 = ?, V2 = 20$$
$$4 \times 100 = C2 \times 20$$
$$C2 = \frac{4 \times 100}{20} = \frac{400}{20} = 20M$$
$$1X : 1Y$$
$$4M : \mathbf{20M}$$

20. If 20 ml of compound X in a 2M solution is required to react completely with 4 ml of compound Y, what is the strength of compounds Y's solution, if the reaction is 1X:1Y?

$$C1 = 2, V1 = 20, C2 = ?, V2 = 4$$
$$2 \times 20 = C2 \times 4$$
$$C2 = \frac{2 \times 20}{4} = \frac{40}{4} = 10M$$
$$1X : 1Y$$
$$2M : \mathbf{10M}$$

21. If 100 ml of compound X in a 5M solution is required to react completely with 20 ml of compound Y, what is the strength of compound Y's solution if:

$$C1 = 5, V1 = 100, C2 = ?, V2 = 20$$
$$5 \times 100 = C2 \times 20$$
$$C2 = \frac{5 \times 100}{20} = \frac{500}{20} = 25M$$

a. The reaction is 1X:1Y

$$1X : 1Y$$

$$4M: \mathbf{25M}$$

b. The reaction is 1X:2Y

$$1X: 2Y$$
$$4M: \mathbf{50M}$$

c. The reaction is 1X:5Y

$$1X: 5Y$$
$$4M: \mathbf{125M}$$

d. The reaction is 2X:5Y

$$2X: 5Y = \ 1X: 2.5Y = 4M: \mathbf{62.5M}$$

e. The reaction is 1X:0.5Y?

$$1X: 0.5Y = 4M: \mathbf{12.5M}$$

Email: ask.let@uea.ac.uk